

Zero waste realization method of rice husk: Silica extraction and methylene blue adsorption

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Highlights

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The optimal conditions for extracting Si from rice husk (RH) were established.

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Si-extracted RH (Si-RH) was effective in the adsorption of methylene blue (MB).

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The adsorption of MB by Si-RH occurs by a complex mechanism.

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Si-RH can be stably recycled as an adsorbent.

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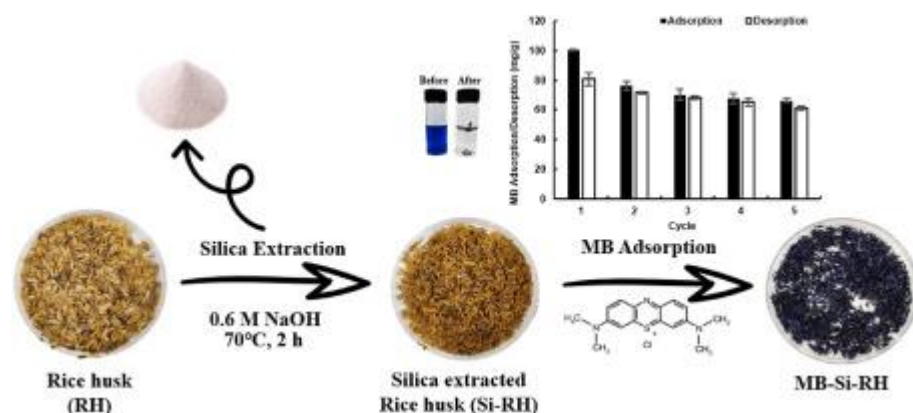
The Si extraction and dye adsorption process is a zero-waste utilization technology of RH.

Abstract

This study determined the optimal conditions for extracting Si from rice husk (RH) for efficient resource utilization of RH and evaluated the adsorption characteristics of methylene blue (MB) by Si-extracted RH (Si-RH). The optimal conditions for extracting silica from RH were mixing 0.6 M NaOH (1:10 ratio) with RH (>2 mm) and reacting at 70 °C for 2 h. The maximum adsorption amount of MB by Si-RH with expanded surface area and reinforced cellulose-hemicellulose-lignin matrix was 99.01 mg/g, which was three times higher than that of RH (34.36 mg/g), and better fit the Langmuir isotherm and Pseudo-second order models. It was found that the adsorption of MB by Si-RH was greatly affected

by environmental changes such as reaction time, pH, dosage, and reaction temperature. The surface properties by SEM-EDS and FTIR and the adsorption model equation showed that the adsorption of MB by Si-RH was dominated by complex mechanisms rather than a single mechanism, and these greatly influenced the initial adsorption-desorption process of MB by Si-RH. In continuous adsorption-desorption experiments, Si-RH showed excellent adsorption efficiency for MB. In conclusion, the extraction of silica from RH and utilization of the residue as a dye adsorbent proposed in this study is considered to be the optimal method for realizing zero waste of RH.

Graphical abstract



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Introduction

Rice is one of the major food sources in Asia, producing about 92 % of the world's rice production. Typically, RH is composed of 50 % cellulose and polysaccharides such as hemicellulose, 30 % lignin, and 20 % inorganic components, of which more than 90 % is amorphous silica (Singh, 2018). Unlike other biomass, RH have the characteristic of containing high silica, which makes them less useful as general biomass resources such as energy sources or feed (Kim et al., 2017), and they have the disadvantage of being difficult to corrode because the outer skin is densely covered with silicon. Therefore, research is being continuously conducted to separate and extract silica present in RH and utilize it as a high-purity silicon raw material, cement additive, catalyst, cosmetic additive, etc. (Chun et al., 2020; Song et al., 2018; Zulkifli et al., 2013). Extracting silica from RH is low cost because it uses agricultural waste, and it has the potential to produce high value-added materials, so it is attracting attention in modern industries that value sustainability (Ali et al., 2023; Nzereogu et al., 2023; Tariq et al., 2024). In particular, RH is generated during the rice milling process, and since it is stored in an unmilled state and milled when needed, a

certain amount is constantly generated every year, making it one of the materials with very high industrial resource value (Quispe et al., 2017). The amount of RH produced varies depending on the rice variety, cultivation environment, and milling process, but it is generally estimated that about 20 % of the weight of rice is produced, and that about 80 million tons of RH are produced annually worldwide (Armesto et al., 2002). For this reason, extracting silica from RH is a groundbreaking technology that can not only reduce agricultural waste but also develop high value-added new materials.

Combustion and chemical treatment methods are commonly used to extract silica from RH. Among the components of RH, carbohydrates are known to be removed by combustion at 200–300 °C, and lignin is known to be combusted at a temperature of over 575 °C (Satbaev et al., 2021). Therefore, the method of obtaining silica by burning RH at high temperature is the simplest method, but it has the problem of causing greenhouse gas emissions and air pollution (Zulkifli et al., 2013). In addition, RH ash has the disadvantage of not being able to obtain high-purity silica because other alkali impurities are mixed in the silica (Hamidu et al., 2025). Extraction of Si through chemical treatment of RH is known to be a more effective method than the conventional combustion process due to its efficiency in greenhouse gas emissions, cost, and time required. When silica is extracted under alkaline conditions, lignin can be dissolved by inducing the separation of ester rings in biomass, and residual organic compounds contained in the silica extract can be removed through thermal decomposition at temperatures above 575 °C (Chun and Lee, 2020).

However, the resource utilization plan for RH from which Si is extracted through alkaline treatment is still insufficient, and they are treated as general waste. Since the amount of RH discharged throughout the year is constant, it is urgent to devise a resource utilization plan for them. In particular, establishing a resource utilization plan for these will be a decisive factor in realizing high value-added industries and zero waste through Si extracted from RH in countries that utilize rice as a major food resource.

It was hypothesized that after Si is extracted from RH through alkaline treatment, the lignin and hemicellulose present inside it are partially decomposed and the surface functional groups are changed, which will have a positive effect on the adsorption of pollutants. In general, after extracting Si from woody biomass, the lignin-cellulose-hemicellulose matrix in the residue is known to be further strengthened and its surface area and pore structure are well developed, thereby increasing the adsorption affinity of pollutants (Chen et al., 2021). Several researchers have reported that the adsorption efficiency of various contaminants in alkali-treated biosorbents is improved compared to the raw material (Khan and Shanableh, 2022; Zafar et al., 2020; Almanassra et al., 2022; Ashrafi et al., 2016). However, information on the adsorption of pollutants on RH from which Si is

extracted is limited, and systematic studies related to adsorption-desorption are needed. In particular, most of the existing reported studies have been conducted separately on the extraction of Si from RH and the adsorption characteristics of alkali-treated biosorbents for pollutants, which limits the ability to secure a zero-waste resource utilization plan for RH (Chun and Lee, 2020; Nzereogu et al., 2023; Khan and Shanableh, 2022; Zafar et al., 2020).

Therefore, the purpose of this study was to (i) identify the optimal conditions for extracting Si from RH by alkaline treatment and (ii) evaluate the adsorption-desorption characteristics of MB using the residue after extraction in order to provide information for realizing zero waste recycling of RH.

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Section snippets

Chemical reagents

Reagents such as HCl (Samchun Chemical Co., Korea), NaOH (Samchun Chemical Co., Korea), HClO₄ (Daejung Chemicals, Korea), H₂SO₄ (Daejung Chemicals, Korea), methanol (Samchun Chemical Co., Korea), and MB (Samchun Chemical Co., Korea) used in this experiment were of analytical grade and were diluted to suit each designed experimental condition.

Optimal conditions for extracting Si from rice husk

Prior to this experiment, RH was classified into particle sizes determined by sieving, and the samples were sealed and stored. To determine the optimal

Optimal conditions for extracting silica from rice husk

The silica extraction yield according to particle size is shown in Fig. 1A. The silica extraction efficiency was 23.4, 32.7 and 41.4 % for particle sizes of <0.2 mm, 0.2–2.0 mm, and >2 mm, and the extraction efficiency of silica from RH decreased as the particles became smaller due to grinding. This result is because the absorption of the solution increased as the specific surface area of the RH increased due to grinding, and thus the amount of the extracted liquid relatively decreased. Gu et

Conclusion

This study was conducted to efficiently recycle the residue remaining after extracting Si from RH. To establish the optimal conditions for extracting Si from RH, experiments were conducted under different conditions of RH particle size, NaOH concentration, maxing

ratio of RH and NaOH (w/v), reaction time, and reaction temperature. From our results, the optimal conditions for extracting Si from RH were designed to be a mixing ratio of 0.6 M NaOH and unpulverized RH of 1:10, and extraction in a

CRediT authorship contribution statement

Ye-Ji Lee: Writing – original draft, Data curation. **Jeong-Min Lee:** Investigation. **Jung-Won Park:** Investigation. **Jim J. Wang:** Writing – review & editing. **Meng Wang:** Investigation. **Dong-Cheol Seo:** Writing – review & editing. **Jong-Hwan Park:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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